

A primer on Testing of Hypotheses

UGAIN – Black Belt in Six Sigma – Module 3

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Additional information is available on the notes pages.

We recommend to consult the professional literature for those who really need to apply hypotheses testing in practice. Some more information can also be found in the document ‘Black Belt in Six Sigma – Testing of Hypotheses – Further Reading (Dutch).pdf’.

Agenda and tentative timing

Subject		Tentative timing (min)
Case example – acceptance sampling		15'
The inference procedure explained		45'
The concept p-value		10'
Pitfalls		10'
Examples in SPC		40'
		120'

Single sample acceptance sampling by attributes formulated as a statistical hypothesis test

Case example

- Lot: $N = 10000$ expensive ceramic tiles.
 - AQL = 2% (i.e., number of defectives $D \leq 200$)
 - Acceptance sampling based on $(n=100, c)$
 - Statistic = d = number of defectives in SRS
 - If $d \leq c$: accept, else reject.
 - What value of c is appropriate?
- As long as the entire lot has not been scrutinized, any statement about the quality of the lot remains a **hypothesis**.
 - Within the framework of statistical testing, **two mutually exclusive hypotheses** are formally put forward:
 - H_0 : the lot is acceptable, i.e., $P = D/N \leq \text{AQL}$
 - H_A : the lot is not acceptable, i.e., $P > \text{AQL}$ (*)
 - H_0 expresses the **status quo**, H_A expresses the situation requiring **special action**.
 - Suppose in the sample of 100 tiles, $d = 4$ defectives are found (i.e., $p = 4/100 = 4\% \gg \text{AQL}$). **Which of the hypotheses should we accept?**

* Also referred to as H_1 .

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In case 4 defectives are found in the sample, this is what the stakeholders (the producer and the consumer) might conclude:

- Producer: if AQL = 2%, then there can be up to 200 defective tiles in the lot, so it is perfectly possible that 4 defectives appear in a random sample of size 100 drawn from an acceptable lot.
- Consumer: based on the sample, we estimate the percentage defectives to be 4%, this is double the value of AQL. This strongly suggests that the lot is not acceptable for further use.

Statistical testing: potential errors and their risks

- The procedure is not guaranteed to be error-free (see on the right).
- The procedure is not deterministic, as based on a SRS. Repetition of the procedure may lead to a different conclusion. Hence the **errors are stochastic in nature, with a risk of occurrence.**
- The errors are **mutually exclusive**. But as we don't know which error will occur, we are obliged to **cover ourselves against both.**
- Luckily, the error risks are **calculable** and **controllable**.
- Given the fact that H_0 expresses the status quo, **α should definitively be small.** But what about β ?
- Involved parties should **agree on the risks of the procedure up-front.**

		Reality	
		H_0 true	H_A true
Our call	H_0 true	OK	Type-II error Risk = β
	H_A true	Type-I error Risk = α	OK

In the context of acceptance sampling, α and β have this specific interpretation:

α = the producer's risk,
 β = the consumer's risk.

The risks are controllable. However this may come at a cost that we are not always willing to take in practice.

It is important that the hypotheses are carefully chosen. H_0 always has to express the status quo, that is the situation where things can go their normal way (aka the do-nothing scenario). If we reject H_0 (and accept H_A), we state that special action will be required. So we cannot switch the hypotheses: H_0 cannot express that the lot is bad.

Some examples:

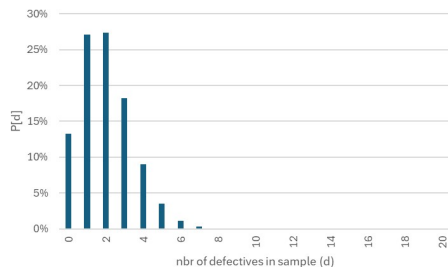
- **A lot of tiles:** normally the lot is OK for further use. However when we accept H_A , we need to take corrective action. This will likely interrupt the construction works.
- **A lawsuit (e.g., a murder trial):** H_0 stands for 'the defendant is innocent', will not be punished and can go on with her/his life, whereas H_A would require the imposition of an appropriate punishment.
- **An investment file (e.g., in a new technology):** here H_0 represents the case that the new technology will not be better than the current (proven) one. Accepting H_0 means literally do nothing, don't go on with the investment!

As a consequence we cannot dismiss H_0 lightly. So α needs to be small. The situation for β is more subtle. For instance, accepting a lot of tiles with 2.1% defectives (so strictly speaking a non-acceptable lot), may not be so tragic. However accepting a lot with 10% defectives is really a serious mistake. We will clarify this point when dealing with the OC-curve of the procedure.

Another important point is that the involved parties need to agree on the risks up-front. To be precise: there should be agreement on the OC-curve of the procedure.

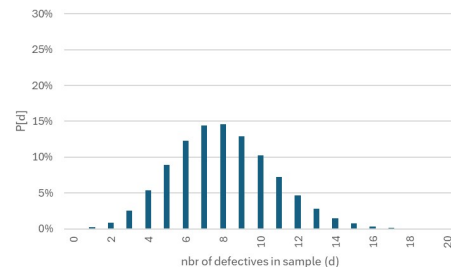
Sampling distributions under H_0 and H_A

Distribution of the statistic d in a lot with $P = 2\%$ defectives (H_0 is valid)



- $P[d \leq 4] \cong 95 \%$

Distribution of the statistic d in a lot with $P = 8\%$ defectives (H_A is valid)



- $P[d \leq 4] \cong 10 \%$

For low values of d , we will accept H_0 . For high values, we will accept H_A .
What is the most adequate trade-off point (critical value = c), to change our mind?

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Since we do not know the true fraction defectives ($P = D/N$) in the lot, the actual outcome of defectives in the sample ($d = 4$), can come from different sampling distributions corresponding to a whole spectrum of potential values for the lot fraction defectives. Evidently P cannot be 0 (because then there would be no defectives in the lot, hence also not in the sample). On the other end of the spectrum $P = 1$ is equally impossible, as then all the tiles would be defective, hence also all the tiles in the sample. But even extreme values of P , like 0.5 % or 90 %, could theoretically lead to 4 defectives in a random sample of size 100. (Indeed, if $P = 0.5 \%$, the number of defectives $D = 50$. If $P = 90 \%$, the number of non-defectives $N - D = 1000$. In both cases 4 defectives out of 100 can occur.

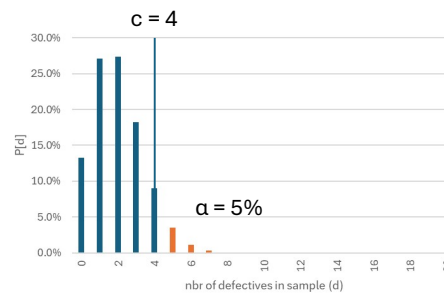
What we see here is that a lot with a limit value of defectives equal to AQL (2 %), will lead to samples of which almost 10 % have 4 defectives. If a lot would have 8 % defectives (4 times the limit of AQL, so already quite bad), the probability of drawing a sample with 4 defectives, drops to 5 %. So 4 defectives in a sample of size 100, points rather in the direction of H_0 , then in the direction of H_A (for a 'sufficiently bad' lot).

The fewer the number of defectives we have in the sample, the more our decision will be in favor of H_0 . Contrary, the higher the number of defectives, the more we will opt for H_A .

Setting the risk α of the type-I error

(The significance - aka the producer's risk)

- We want a good lot to be accepted most of the time.
- Suppose we want a lot with a fraction of defectives equal to AQL, to have an acceptance probability of about 95 % (or more), then we will have to set our critical value at $c = 4$.
- Indeed then $\alpha = P[\text{reject lot}] = P[d > 4] = P[d \geq 5] = 5\%$, as can be seen on the right.
- If we want even smaller values of α , we will have to increase our critical value further. E.g., $c = 5 \Rightarrow \alpha = 1.5\%$.
- α is called the **significance** of the test.



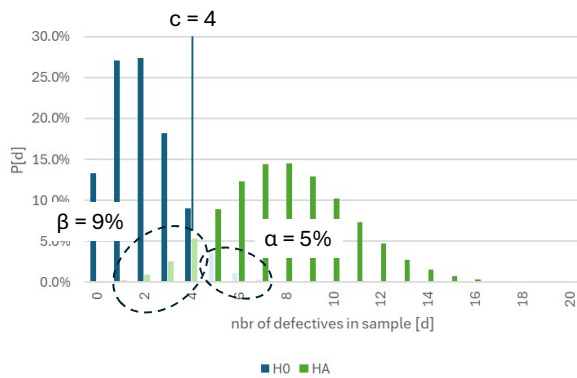
α is 5% for the limit case ($P = \text{AQL}$),
if lot quality is better, α will be smaller.

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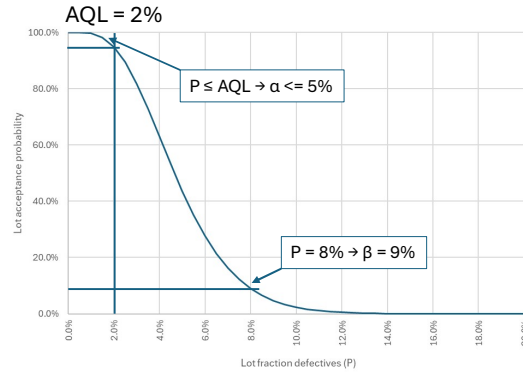
So it is clear that we set α to any value we like. From choosing α (for a given n), β will follow for any given lot fraction defectives P .

The word *significance* has a very precise meaning in the context of statistics. It does not mean important or relevant, as it is often interpreted in common language. The significance of a test is the risk of committing the type-I error, i.e., starting special action where in fact this is not needed or even counter productive. E.g., in the context of process control, Deming demonstrated that adjusting a process that is in fact stable, only increases the variability.

The OC-curve of the procedure



- The critical value c has to be set on the sampling distribution under H_0 ($P \leq AQL$). This is the **reference distribution** (blue color).
- If in reality $P > AQL$ (e.g., 8%), we deduce β from c .

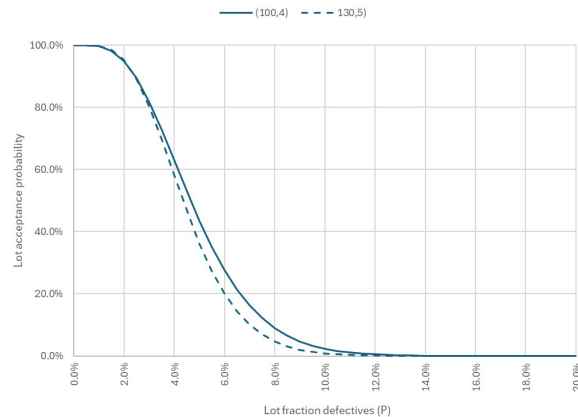


- The graph showing the probability of accepting H_0 for any value of P , is called the Operating Characteristic curve (OC-curve) of the procedure.

The critical value of the procedure has to be set on the sampling distribution assuming the H_0 is true. In reality however, we don't know this. Nevertheless we have no other choice. Once the critical value chosen by setting the value of α , the β -values for any given lot fraction defectives P will follow. The function $P \rightarrow P[d \leq c] = P[\text{accept } H_0]$ is called the OC-curve of the procedure.

We can choose α , but how do we control β ?

- From the OC-curve it is obvious that β gets smaller when P gets larger. So automatically the probability of accepting a bad lot (i.e., committing the type-II error) becomes smaller, whatever value of α we choose. THIS WE GET FOR FREE.
- For a given sample size n , β clearly depends on α . If we want to fully eliminate the risk of ever committing the type-I error, we will have to accept a huge risk of committing the type-II error. Hence setting $\alpha \cong 0$ is not a good practice. THIS WE MUST BE AWARE OF.
- However it is also clear that we cannot expect a small value of β for lots that are just slightly worse than the limit for acceptable quality (i.e., if P is just a bit larger than AQL, then β will be just a bit smaller than $1 - \alpha$. (But this is generally not tragic.) THIS WE HAVE TO ACCEPT.
- We can however ensure that β becomes acceptably small **where it is relevant**, while also maintaining α at a good level. We pay for it in the form of a larger sample size. THIS IS HOW WE CONTROL β (given α).



By increasing the sample size to 130 and setting the critical value to 5, we maintain α at the level of 5%, but reduce β to 5% for bad lots containing 8% defectives.

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Avoiding to ever make the type-I error can be well described with a (kind of) joke: a judge that never sends an innocent defendant to jail, will have to release all defendants, also the guilty ones. In some SPC-control charts, α becomes as small as 0.3% for a single test. However in SPC the process is being monitored on a periodic basis (e.g., every hour, every x parts produced, ...), reducing the risk of not reacting to an out of control state.

Recapitulation – the testing procedure in a nutshell

Step	Description	Illustration
1	Describe the problem in the form of two hypotheses : H_0 and H_A	H_0 : percentage defectives in lot $\leq 2\%$ (lot quality is good) H_A : percentage defectives in lot $> 2\%$ (lot quality is bad)
2	Choose the significance α of the test, this is the probability of wrongly rejecting H_0	$\alpha = 5\%$ There is a probability of 5 % that a good lot will be seen as bad.
3	Determine an appropriate test statistic and its sampling distribution	Test statistic = number of defectives in sample ; sampling distribution = hypergeometric (approximated by binomial)
4	Find the best sampling size, taking into account the error risks and practical constraints	If we want to limit the acceptance probability of a lot with 8% defectives to 5% (while keeping α also to 5%), we need to draw a sample of size 130 (this is 1.3 % of the lot).
5	Determine the reference distribution and the critical value (by measuring the significance on the reference distribution)	The reference distribution is the sampling distribution assuming H_0 holds. Here this is a binomial distribution with parameters $n = 130$ and $\pi = 2\%$. The critical value is 5.
6	Draw the sample from the population and determine the value of the test statistic	130 are drawn at random from the lot (quite a task!) and inspected; suppose 4 are found to be defective
7	Compare the test statistic to the critical value and draw the conclusion	4 is not higher than 5, we state that we cannot reject H_0 at the 5% significance level, hence we accept the lot for further use

one sided
or
two sided

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The test recipe formulated here is **universal** for any simple (i.e., one stage) statistical procedure, regardless of the object of the test. Tests differ from each other only in the choice of the test statistic (in our example the number of defectives in the sample or the percentage of defectives) and, related to this, the sampling distribution of the test statistic and the critical value.

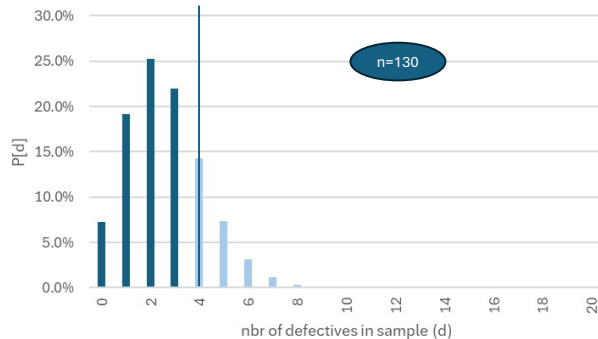
For the sake of completeness: there are one-sided and two-sided tests. Two-sided tests are only used when it comes to detecting differences where it does not matter whether it is 'larger' or 'smaller'. One-sided tests must be adapted to the direction in which a difference is sought.

The most technical aspect is determining the sampling distribution. Here we are dealing with a classic situation: drawing one sample from a finite population. Every statistician then knows that the sample distribution is a so-called hypergeometric distribution (well approximated by a binomial distribution, when the population is not small).

Further in this document, we'll deal with two important tests within the context of process control and hence very relevant for 6σ : the t-test and the χ^2 -test.

The notion of p-value

The p-value of the observed test statistic ($d = 4$).



- The p-value expresses how well the observed value of the test statistic supports the hypothesis H_0 . Technically: p-value = probability of a value that deviates at least as much as the observation, **assuming that H_0 is true.**
- Consequently, **a decision can be made by directly comparing the p-value with the significance α , no need for a determining the critical value anymore.**
- The p-value is standard output of any decent statistical software.
- The advantages of the p-value are: the procedure becomes easier + **everyone chooses their own significance.**

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Like the significance, the p-value is read from the reference distribution. So comparing the test statistic to the critical value is equivalent to comparing the p-value to the significance. Because the p-value is calculated automatically by statistical software, this simplifies the procedure.

The main reason for introducing the p-value, however, is that it allows everyone to use their own significance. It is best practice in statistical research to report the p-value (and not merely whether a significant difference was noted or not). This is extremely important in the context of scientific publications, an argument that we will not discuss in this introductory document.

Within business context, statistical testing should be seen as a tool of **risk management**. Let's consider again the example of an investment file. Remember that in such case H_0 expresses the hypotheses that the investment will not bring value. So accepting H_0 means that the investment will be cancelled. If this would be the wrong choice, the consequence would be that the value the investment would bring after all, would be missed. If the investor is conservative (risk averse), he will apply a small value for α leading to a higher probability that the p-value will be larger than α and so the investment will not go through. However a risk investor will set a larger value for α , and with the same value of the test statistic may well reject H_0 (at the risk of losing his money).

After all α is a design parameter in the procedure and can be set at any value.

For completeness: in a two-sided test, α is 'distributed' over both tails of the distribution, hence the same should be done with the p-value. This is explained in the document 'Black Belt in Six Sigma – Testing of Hypotheses – Further Reading (Dutch).pdf'.

Pitfalls of statistical testing

Set α after knowing the test statistic

- Doing this destroys the entire meaning of the test, as it will again **lead to biased decision making**. The value of α should be said prior to starting the procedure and should be agreed upon between the stakeholders.

After **accepting** H_0 believing it is **true**

- That would mean neglecting the fact that the risk of committing the type-II error is not zero and is even usually quite high if reality is 'close to' H_0 . See for example the OC-curve of the (100, 4)-procedure to test lots of ceramic tiles. A lot with a fraction defectives slightly higher than AQL, will almost always pass the test. (Which is acceptable!)

Believing a **significant** result is always **relevant**

- Continuing the example of the previous point. We could test the lots with very large samples (e.g., $n = 5000$), to create a very steep OC-curve (see the notes page). In this way lots that are even slightly above AQL, will be rejected. This is evidently not a desirable practice. In times of abundant data availability this is a particular challenge (**big data fallacy trap**).

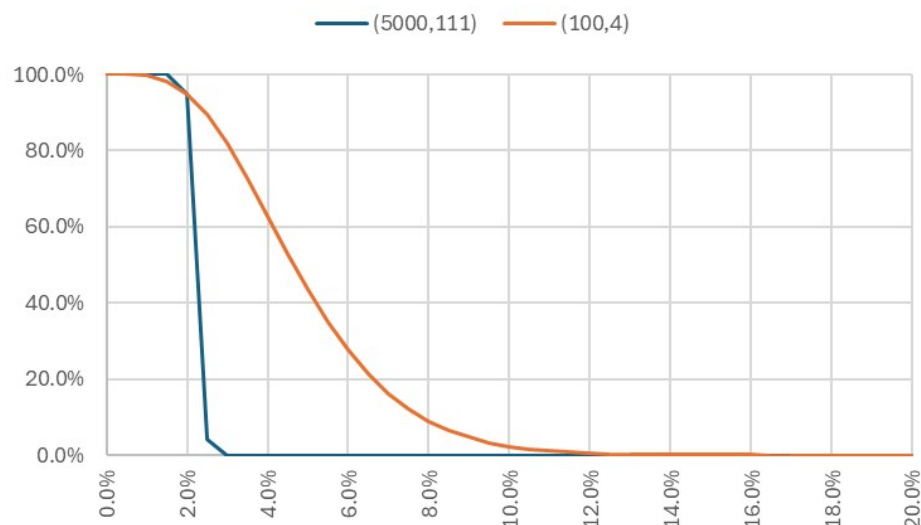
Discarding the **efficiency** argument

- Usually there are several testing procedures with (approximately) the same OC-curve. Statistically speaking these procedures are equivalent. However the OC-curve does not provide any information on the efficiency (workload) of the procedure. As an example: it can be shown that the procedure (175,8) and (90,2,7)+(90,8) are equivalent. The latter is however more efficient.

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The first pitfall is again in particular relevant within the scientific community. For instance a lot of research in the fields of psychology and medicine is based on statistical procedures. Suppose we want to demonstrate that some drug makes a difference, then H_0 expresses the hypotheses 'no difference'. Suppose after carrying out the experiment that the p-value is 9 %. Somebody who eagerly wants to conclude that the drug is effective, may be tempted to set a value of α of 10 % or higher. Evidently leading to a high risk of committing the type-I error.

Testing lots of size 10000 with AQL = 2%. If the sample size becomes very large, small deviations from AQL will already be detected as 'bad quality'.



More on efficiency. The choice of the test statistic plays an important role here. Testing of quality by variables is much more efficient than testing by attributes, as information gets lots when translation the variable measurements into the GO/NOGO-attribute.

The same applies in the case of so-called non-parametric tests (e.g., the Wilcoxon signed ranks test as compared to the t-test). Efficiency is related to the notion of discernment power of a test. See the professional literature for more information on this topic.

Example t-test for the mean

Case description

A manufacturing process produces parts with a critical dimension that is normally distributed around the value of 10.

After replacement of a broken machine component, management suspects that parts have slightly become smaller, increasing the risk of sending bad parts to the customer.

A careful inspection of 20 randomly selected parts, leads to these data :

10.06 – 9.89 – 9.90 – 9.99 – 9.87 –
9.88 – 9.93 – 9.98 – 10.12 – 9.86 –
9.89 – 9.81 – 9.95 – 9.92 – 10.07 –
9.98 – 10.02 – 9.62 – 9.96 – 9.86

Is management's fear justified?

Test procedure

H_0 : the process is still well centered

H_A : the process has shifted down

$\alpha = 2\%$

rather conservative choice, only adjust process if evidence is strong

test statistic = $\frac{\bar{X} - 10}{s/\sqrt{n}} = -2.95$

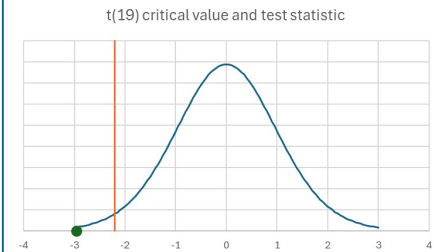
classical quantity for testing the mean of a distribution

reference distribution = $t(19)$

critical value = -2.20

p-value = 0.4%

conclusion: H_0 cannot be accepted at 2% significance, process needs adjustment

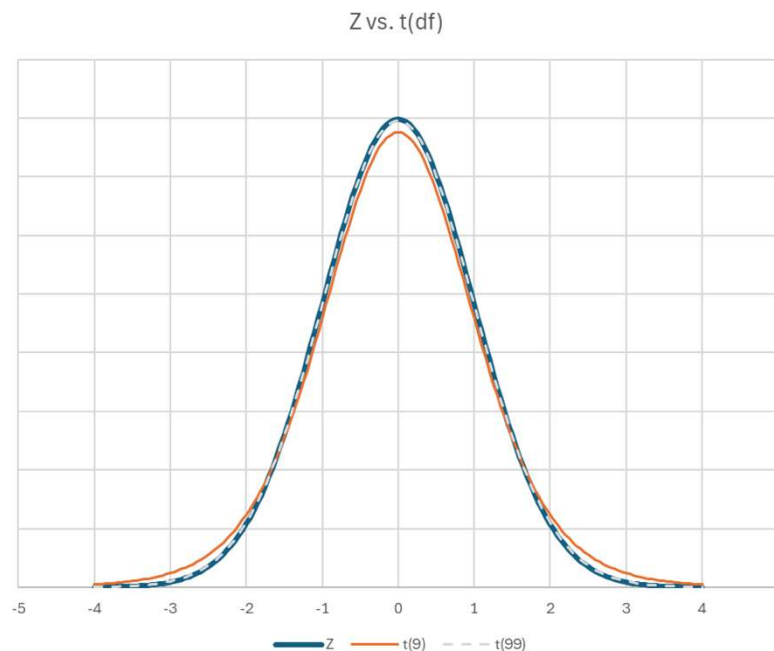


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See the excel-file 'Black Belt in Six Sigma – Testing of Hypotheses.xlsx'.

We know that when a quantity X is normally distributed, with mean μ and standard deviation σ , then the quantity $Z = \frac{X - \mu}{\sigma}$ is normally distributed with mean 0 and standard deviation 1, notation $Z \sim N(0,1)$.

When we have a sample of values (size n) of X , then the value $t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$ follows a so-called t-distribution with $n - 1$ degrees of freedom. The t-distribution is similar to the Z-distribution. It is bell-shaped, symmetrical and has mean 0. Its standard deviation is however somewhat larger than 1. When $n \rightarrow \infty$ then $t(n) \rightarrow Z$.



Example χ^2 -test for the standard deviation

Case description

A manufacturing process produces parts with a critical dimension that is normally distributed around a nominal value with standard deviation of 0.01.

Due to wear of a machine component, management suspects that the process has become slightly more erratic, increasing the risk of sending bad parts to the customer.

A careful inspection of 20 randomly selected parts, leads to these data :
 9.997 – 10.004 – 10.010 – 9.991 – 10.008 –
 9.981 – 9.984 – 9.993 – 9.990 – 9.997 –
 9.996 – 9.977 – 9.999 – 9.981 – 10.000 –
 10.010 – 10.005 – 10.011 – 9.997 – 10.022
 Is management's fear justified?

Test procedure

H_0 : process variability has not changed

H_A : process has become more erratic

$\alpha = 2\%$

rather conservative choice, only adjust process if evidence is strong

test statistic = $\frac{(n-1)s^2}{\sigma^2}$

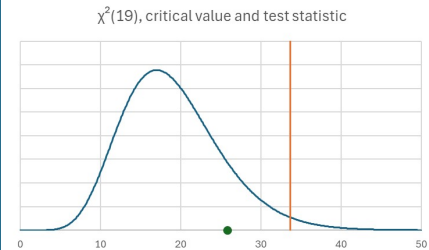
classical quantity for testing the variance of a distribution

reference distribution = $\chi^2(19)$

critical value = 33.69

p-value = 13.6%

conclusion: at 2% significant H_0 cannot be rejected, process variability judged stable



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See the excel-file 'Black Belt in Six Sigma – Testing of Hypotheses.xlsx'.

In SPC the common tests for process variability are the R-test or s-test. Here we apply the χ^2 -test.

When we have a sample of values (size n) of a normally distributed quantity X with standard deviation σ , then the value $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$ follows a so-called χ^2 -distribution with $n - 1$ degrees of freedom. The χ^2 -distribution is an asymmetric distribution taking only positive values. When $n \rightarrow \infty$ then $\chi^2(n) \rightarrow N(n, \sqrt{2n})$.

χ^2 -density functions

